NAZARIO SALDANA

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EDUCATION

The University of Texas at Austin – Austin, TX **May 2028**

Degree: Bachelor of Science in Electrical & Computer Engineering

Relevant Coursework: Digital Logic Design, Embedded Systems, Software Design and Implementation, Circuit Theory

PROJECTS

Ascend ATV– *Embedded Firmware* **September 2025 - Current**

- Delivered fully functional dual-linear actuator system after verifying voltage and response on integrated software
- Configured RaspberryPi GPIO mapping to ensure accurate actuator control and system performance
- Introduced detailed wiring schematic and load cell documentation to streamline debugging of linear actuators
- Migrated linear actuator operation from Raspberry Pi to NVIDIA Jetson Nano to boost system performance
- Enhanced control software to synchronize linear actuator operation and enable live feedback response

Space Invaders Game Development – *C, Embedded Systems* **November 2025**

- Built a Space Invaders game on the MSPM0G3507 using an interrupt system for real-time input, graphics, and audio
- Implemented low-level drivers for ADC slide-pot control, GPIO buttons, SysTick timing, and 12-bit DAC playback
- Integrated the ST7735 SPI LCD with optimized sprite rendering, background drawing, and efficient redraw routines
- Used UART for debugging, telemetry output, and validating driver behavior during system integration
- Designed modular firmware including collision detection, object state machines, and event timing

Custom Stopwatch/Timer Processor – *Verilog, FPGA* **November 2025**

- Designed a 4-mode digital counter supporting count-up, count-down, and preset load operations across 00.00–99.99
- Implemented a 7-state HLSM in behavioral Verilog to control system logic and mode transitions
- Built two clock dividers using 16-bit and 20-bit counters to drive logic timing and display multiplexing frequencies
- Developed BCD arithmetic logic with 4-bit overflow/underflow detection for accurate decimal counting
- Synthesized and validated the design on a Basys3 FPGA, achieving 100% accuracy during hardware testing

Dungeon Crawler RPG Development – *C++* **April 2026**

- Engineered a C++ application utilizing deep class hierarchies, inheritance, virtual functions, and constructor chaining
- Managed dynamic object ownership across inventories and loot tables, via custom destructors to prevent memory leaks
- Validated dynamic memory allocation and pointer safety by achieving zero leaks through Valgrind on Linux servers
- Utilized dynamic_cast and C++ Standard Template Library (STL) vectors to securely process type-safe runtime logic

Learning Animal Guessing Game – *C* **March 2026**

- Developed an animal-guessing game using binary decision trees and iterative traversal with explicit stacks
- Implemented a hash table for stats and an undo/redo system with dual stacks to manage tree memory
- Developed BFS-based validation logic to verify tree integrity and ensure no cycles or unreachable nodes

LEADERSHIP & COMMUNITY INVOLVEMENT

Society of Hispanic Professional Engineers – Austin, TX **August 2024 - Current**

Recruitment & Retainment Co-Chair

- Organized upwards of 15 events, including a themed cookout with food and games to increase interaction
- Boosted membership by 25% by fostering an inclusive environment through outreach and personal engagement
- Led the Manitos & Manitas mentorship program, including pairings and events of 100+ members
- Managed game logistics for 100+ members, including partner pairings, target data, and real-time communications

Ramshorn Mentor – Austin, TX **August 2025 – Current**

Academic Coach

- Mentored four first-year Ramshorn Scholars, supporting academic, social, and personal transition into college
- Tutored in challenging subjects to strengthen academic performance and develop effective study strategies
- Supported professional growth via career resources, reviewing resumes, and facilitating networking opportunities
- Introduced Review-A-Thon to help increase mentee grades by 10% for Electrical and Computer Engineering exams

TECHNICAL SKILLS

- **Languages:** C/C++, ARM Assembly, Verilog, Python
- **Software:** Xilinx Vivado, KiCad, LTspice, AutoCAD, Autodesk Inventor, Revit
- **Technical Skills:** PCB Design, Oscilloscope, Circuit Simulation, SMD/SMT Soldering, 3D Modeling, 3D Printing

HONORS

Ramshorn Scholar | HSF Scholar | University Leadership Network Scholar | Cockrell Engineering Scholarship Recipient